3rd Project : Student Performance Prediction Dashboard

Student Performance Prediction Dashboard – Recommendations & Logic Behind the Recommendation

1. Home

- Recommendation: Read the project overview before using the dashboard.
- Logic Behind the Recommendation: The Home page explains the project's objective, workflow, key features, and how the prediction system should be used.

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2. Dataset Overview

- Recommendation: Ensure the dataset is complete and correctly structured before making predictions.
- Logic Behind the Recommendation: The Dataset Overview page displays dataset shape, feature descriptions, data types, missing values, and sample records to verify data quality.

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3. EDA Dashboard

- Recommendation: Focus on improving the factors that show a strong relationship with student success (e.g., study hours, attendance, previous grades).
- Logic Behind the Recommendation: The EDA Dashboard visualizes distributions, correlations, and feature relationships, helping identify which variables most influence performance.

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4. Prediction

- Recommendation: Increase study hours, maintain high attendance, complete assignments, and improve time management if the model predicts a risk of failing.
- Logic Behind the Recommendation: The Prediction page uses the trained Logistic Regression model to estimate pass/fail probability and generates personalized recommendations based on the entered student information.

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5. Model Performance

- Recommendation: Trust the prediction only after reviewing the model's evaluation metrics.
- Logic Behind the Recommendation: The Model Performance page presents Accuracy, Precision, Recall, F1-Score, ROC Curve, Precision–Recall Curve, Confusion Matrix, and Classification Report to demonstrate model reliability.

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6. About

- Recommendation: Understand the project methodology and technologies before deployment or presentation.
- Logic Behind the Recommendation: The About page summarizes the project architecture, machine learning model, development tools, author information, and overall implementation details.